

Optimizing Mobile Game UI Usability through HCI Evaluation Methods and Emerging Technologies

Iman Saufik Suasana

Department of Computer System
Universitas Sains dan Teknologi Komputer
Semarang, Indonesia
iman.saufik@stekom.ac.id

Budi Hartono

Department of Informatic Engineering
Universitas Sains dan Teknologi Komputer
Semarang, Indonesia
budi@stekom.ac.id

Andik Prakasa Hadi

Department of Informatic Engineering
Universitas Sains dan Teknologi Komputer
Semarang, Indonesia
andikprakasa.hadi@stekom.ac.id

Rudjiono

Department of Informatic Engineering
Universitas Sains dan Teknologi Komputer
Semarang, Indonesia
rudjiono@stekom.ac.id

Abstract—The increasing complexity and popularity of mobile games have underscored the critical importance of User Interface (UI) design in enhancing player experience. As mobile gaming evolves, ensuring an intuitive and efficient UI has become a fundamental challenge for developers. This study aims to explore the effectiveness of Human Computer Interaction (HCI) evaluation methods in optimizing mobile game UI usability, with a particular focus on heuristic evaluation, cognitive walkthroughs, and usability testing. A systematic review methodology was employed to analyze and synthesize existing literature on these methods applied to mobile game UI. The study reveals that while each HCI method offers valuable insights, combining them leads to a more comprehensive understanding of usability issues, addressing both general flaws and context-specific challenges. Key findings suggest that heuristic evaluations are effective in identifying broad usability concerns, cognitive walkthroughs excel in task-oriented games with complex narratives, and usability testing provides the most detailed feedback on user satisfaction. Furthermore, the study highlights the need for adaptive usability evaluation methods in response to the growing integration of emerging technologies such as artificial intelligence (AI) and augmented reality (AR) in mobile games. The research concludes by recommending the integration of these methods into the early stages of mobile game development to enhance user experience and reduce abandonment rates, while also suggesting areas for future research in adaptive usability evaluation techniques for next-generation gaming technologies.

Keywords—Usability Evaluation, Mobile Games, HCI, Heuristic Evaluation, Mobile UI

I. INTRODUCTION

In recent years, the mobile gaming industry has expanded rapidly, with millions of users worldwide and growing demand for optimal user experiences. A key factor influencing this experience is the design of mobile game User Interface (UI), which directly affects how players interact with the game [1]. Human-Computer Interaction (HCI) has played a central role in improving these interfaces by providing usability evaluation methods such as heuristic evaluation, cognitive walkthroughs, and usability testing [2], [3]. These methods are essential for

identifying usability issues and refining UI designs to ensure effective and satisfying user interactions.

However, a significant gap remains in the practical application of these methods within the mobile gaming industry. Many developers struggle to incorporate usability testing effectively into the development process, often leading to complex games with poor usability and high abandonment rates. Prior studies have generally focused on individual methods [4]–[6], with limited comparative or synthesized insights on their use in mobile game UI. This study addresses that gap through a systematic review of existing research, offering a clearer understanding of each method's strengths and limitations, as well as actionable recommendations for their integration into mobile game design [7], [8].

The relevance of this research goes beyond theoretical contributions, it has direct implications for improving user retention and satisfaction in an increasingly competitive mobile gaming market. By synthesizing insights from previous studies, this paper presents a comprehensive evaluation of usability methods and emphasizes the need for their adaptation as technologies like AI and AR reshape mobile gameplay [9]. In doing so, it contributes to the field of HCI by proposing a methodological framework that supports innovation in usability assessment, aligning with advances in agile development, interactive system architecture, and AI-integrated game design.

This study is guided by the following research questions (RQs):

- RQ1: What are the comparative strengths and limitations of heuristic evaluation, cognitive walkthrough, and usability testing when applied to mobile game UI?
- RQ2: How do different usability evaluation methods align with game genres and UI complexity?
- RQ3: What are the emerging challenges in usability evaluation methods when dealing with AI- and AR-based mobile games?

II. LITERATURE REVIEW

A. Human-Computer Interaction (HCI) and Mobile Game UI Design

Human-Computer Interaction (HCI) focuses on the design and evaluation of interactive systems, aiming to optimize the user experience by making interfaces more intuitive and efficient [10]. In mobile game design, HCI plays a crucial role in ensuring that User Interface (UI) are user-friendly, easy to navigate, and visually engaging. The complexity of mobile games, combined with the limited screen size and unique user interactions (touch, gestures), presents specific challenges that HCI methodologies can address. Early studies focused on the general application of HCI to interactive systems, but recent research has shifted towards its application in mobile environments, with a particular focus on mobile gaming [11]–[13]. For example heuristics have been widely adopted to evaluate usability in mobile game UI, helping developers identify usability flaws that affect gameplay and player engagement. More recent works have expanded these methods to account for mobile-specific considerations like touch interfaces and multi-touch gestures [14].

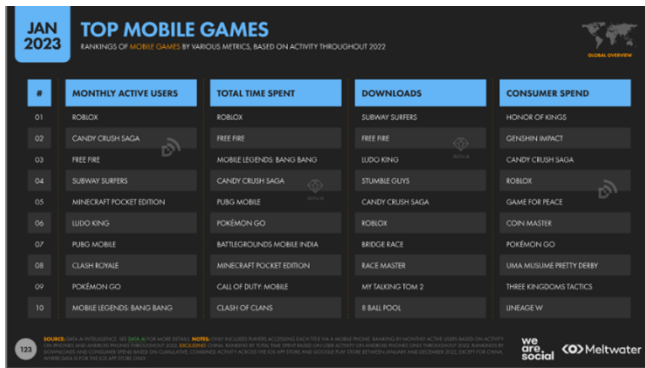


Fig. 1. Top Mobile Games in January 2023 based on various metrics, including monthly active users and total time spent

To illustrate the complexity of game design and how these aspects influence the mobile gaming experience, Figure 2 highlights key aspects for game designers, which are essential for creating an engaging and effective game UI [15]. These aspects include communication, psychology, technical writing, and design principles, each playing a pivotal role in ensuring usability. Additionally, listening skills, as emphasized in Figure 1, reflect the importance of user feedback in mobile game design, where designers must adapt UI based on player preferences and usability challenges [16].

B. Usability Evaluation Methods in HCI

Usability evaluation methods in HCI are designed to assess the ease with which users can complete tasks using an interface. These methods include heuristic evaluation, cognitive walkthroughs, and usability testing. Heuristic evaluation, introduced by [17], involves reviewing an interface against established usability principles to identify potential usability issues. Cognitive walkthroughs, on the other hand, involve

simulating the user thought process as they interact with the system to identify usability barriers at each step of the user journey. Usability testing, which involves observing real users interacting with the game, provides more comprehensive feedback on user performance and satisfaction [18]. Each method has its strengths, heuristic evaluation is efficient and cost effective but may overlook context-specific issues, cognitive walkthroughs provide detailed insights into task complexity but are time-consuming, usability testing offers in-depth, context-rich data but requires substantial resources. Previous studies have compared these methods, revealing that no single method is universally superior, and that a combination of methods tends to provide the most reliable results [19], [20].

While prior research predominantly applies isolated evaluation techniques, few studies have engaged in systematic triangulation across methods. Moreover, the theoretical underpinnings that justify method selection, especially in genre-specific or AI-enhanced environments remain underexplored. This study builds on the intersection of interaction design theory and evaluative heuristics to address these deficiencies [21].

TABLE I
GENRE-SPECIFIC USABILITY EVALUATION METHODS FOR MOBILE GAME UI

Genre Game	Description
Action	The action genre in video games can be further categorized into two primary sub-genres: first-person shooters and third-person games. In first-person games, players perceive the game world from the perspective of the in game character, as if they are viewing it through the character eyes. Conversely, in third-person games, players control a visible avatar within the game world, allowing them to see the character they're guiding. These sub-genres are defined by borrowing cinematic terminology and drawing from literary narrative definitions. There is an underlying connection between these two sub-genres. Regardless of whether it a first-person or third person perspective, the player needs a physical link to fully immerse themselves in the virtual gaming world. This connection bridges the player viewpoint and their physical actions in the game world. Action games often require players to perform challenging actions to enhance the gaming experience. In many action games, players must physically execute the desired actions through proper inputs, whereas in other video game genres, players simply select actions, and the computer carries them out [21].
Adventure	Adventure games are defined by their gameplay style rather than their storyline or content. They typically involve solving puzzles through interactions with characters and the game environment, often without reflex challenges or intense action. This makes them uniquely appealing to those who aren't typically drawn to video games due to their minimal emphasis on action-based challenges or time constraints [22].
Puzzle	This genre relies on puzzle-solving and creative thinking [23].
Horror	This genre focuses on creating tension and fear [24].
Spring	This is a genre that includes games developed by independent developers. They often emphasize creativity and innovation [25].

Educational Games	Educational games prioritize learning and aim to teach and reinforce knowledge and concepts. They form a distinct genre due to their primary educational focus, often involving text-based exercises like fill-in-the-blanks and multiple-choice questions. These games can be creatively adapted into engaging formats, such as game shows with educational questions and points, or transformed into adventure or role-playing games where students play the main character and must answer history-related questions or address historical situations accurately [26].
RPG (Role-Playing-Game)	RPG games offer vast, exploratory worlds where groups of players journey through landscapes in pursuit of treasures, coveted items, and means to improve their character experience and health. Their objective is to overcome monsters and challenges encountered along the way [27].
MMORPG	Massively Multiplayer Online Role-Playing Games (MMORPGs) have many groups, each with different missions and objectives. In an RPG, there a clear objective, and after hours of gameplay, it eventually reaches an end. Conversely, in an MMORPG, there might not be a specific conclusion, and players can continue until they've completed all missions or until other MMORPGs or online games demand their time and attention [28].
Simulation	The simulation genre in video games includes a wide range of games that mimic sports, aviation, driving, and the dynamics of urban and small-town life. Simulations are immersive and offer a realistic experience. They serve as practical and cost-effective ways to navigate and recreate training scenarios that would be dangerous and expensive to undertake in the real world [29].
Strategy	Strategy video games demand careful planning and strategic thinking for success. What sets them apart is that designers set rules and objectives, while players determine the strategies needed to achieve those goals and defeat opponents. War games, a subset of strategy games, often simulate real or fictional events and can be real-time or turn-based. Real-time strategy games involve continuous decision-making, resource management, base construction, and unit production. Turn-based strategy games focus on realistic tactics and provide analysis time before taking actions, with some limiting the number of actions per turn [30].
Sport	This game simulates real-world or fantasy sports [31].
Music	This genre combines music elements with player interaction [32].
VR (Virtual reality)	This genre is designed for use in virtual reality environments and utilizes VR technology [33].
AR (Augmented Reality)	This genre combines gaming elements with the real world through technology [34].

For example, Table I compares different game genres and their usability requirements. This table shows that action and adventure genres typically require different usability approaches, which aligns with the evaluation methods discussed. The table emphasizes that usability testing might be more effective for action games due to the real-time performance requirements, while cognitive walkthroughs might be more appropriate for adventure games where complex narratives and puzzle-solving dominate the gameplay.

C. Previous Research on Usability in Mobile Game UI

Several studies have explored the application of usability evaluation methods specifically in mobile game UI. The ap-

plied heuristic evaluation to identify common usability problems in mobile role-playing games, finding that poor feedback and confusing navigation were the most frequent issues. [35] conducted a usability test of a mobile puzzle game, discovering that players experienced frustration when the controls were not intuitively placed. These studies underscore the importance of usability evaluation in the development of mobile game interfaces, particularly as mobile games become more complex and user expectations grow. However, while these studies highlight key usability issues, there is still a lack of systematic reviews comparing the effectiveness of different usability evaluation methods specifically within the context of mobile game UI, a gap this paper seeks to address.

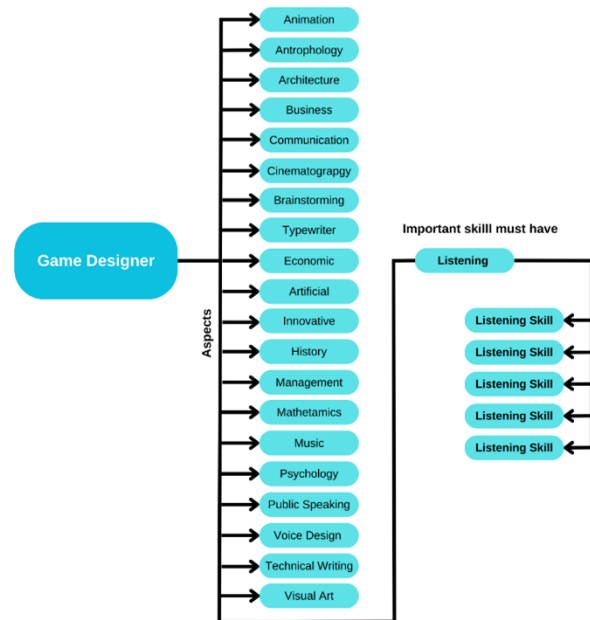


Fig. 2. Key Aspects for Game Designers and Essential Listening Skills

As seen in Figure 2, which ranks the top mobile games by various metrics such as monthly active users and total time spent, the diversity in game popularity and playstyles further underscores the need for different usability evaluation techniques. Popular games like Roblox and Candy Crush have distinct UI that must be evaluated using methods that account for their different player bases and game genres. For instance, games with simpler controls like Subway Surfers might benefit more from heuristic evaluation, while more complex games like PUBG Mobile could benefit from more thorough usability testing to ensure player engagement.

D. Emerging Technologies and Usability Evaluation

As the mobile gaming industry evolves, new technologies such as Augmented Reality (AR) and Artificial Intelligence (AI) are changing the way users interact with games. These technologies introduce new challenges in usability evaluation, as traditional methods may not be equipped to assess complex, dynamic user interactions. For example, AR-based games

require a hybrid interaction model that combines physical and digital worlds, demanding new usability testing methods to evaluate user comfort and interaction flow [36]. Similarly, AI-driven mobile games, which adapt to user behavior, present a challenge for traditional usability methods, as these systems require continuous evaluation to ensure remain user-friendly [37], [38]. Emerging research suggests that existing methods need to be adapted or even entirely new methods developed to evaluate usability in these innovative contexts. This trend highlights the growing need for usability evaluation methods that are flexible and can accommodate the rapidly changing technological landscape of mobile gaming.

E. Research Gaps and the Need for Further Study

Despite the wealth of studies on usability evaluation methods, few have systematically compared these methods within the context of mobile game UI. Moreover, while much of the existing research as [39], [40] focuses on traditional game mechanics, less attention has been given to newer trends like AR and AI in mobile games. This gap in literature emphasizes the need for a comprehensive review that not only examines the usability evaluation methods used in traditional mobile games but also considers the unique challenges posed by emerging technologies. By addressing this gap, the present study aims to provide a more thorough understanding of the usability evaluation landscape in mobile games, with a focus on methods that can be applied to both traditional and cutting-edge mobile game designs.

III. METHODOLOGY

This study adopts a systematic review approach to evaluate and synthesize the usability evaluation methods used in Human-Computer Interaction (HCI) research for mobile game User Interface (UI). The systematic review method was chosen due to its rigorous and transparent approach to identifying, evaluating, and synthesizing research evidence. This methodology allows for the identification of trends, gaps, and strengths within existing usability evaluation methods in the mobile gaming industry.

A. Research Process

The research process for this systematic review is structured into three main phases: search strategy, inclusion and exclusion criteria, and data extraction.

B. Search Strategy

A comprehensive search of peer-reviewed literature was conducted across multiple academic databases, including IEEE Xplore, Google Scholar, ACM Digital Library, and Scopus. The search terms used included combinations of the following keywords: “usability evaluation”, “mobile games”, “HCI methods”, “heuristic evaluation”, “cognitive walkthrough”, and “usability testing”. The search was limited to studies published in the last ten years (2011–2021) to ensure that the review reflects the most recent developments and trends in the field of mobile game usability.

C. Inclusion and Exclusion Criteria

This studies included in this review the following criteria:

- The study focuses on usability evaluation methods applied to mobile game UI.
- The study used at least one HCI method (heuristic evaluation, cognitive walkthroughs, or usability testing) for evaluating mobile game interfaces.
- The study was published in a peer-reviewed journal or conference proceeding.

Studies were excluded if:

- The focus was on mobile applications or non-game UI.
- The usability evaluation was not directly related to HCI methods or was not applied to mobile games.
- The study was a non-peer-reviewed source or grey literature (e.g., reports, preprints).

D. Data Extraction

Data extraction was performed by reviewing the selected studies and recording key details, including:

- The HCI methods used (heuristic evaluation, cognitive walkthrough, usability testing, etc.)
- The type of mobile game UI evaluated (e.g., role-playing game, puzzle game, action game).
- The evaluation outcomes or findings regarding usability issues.
- The context of the study, including the target user group and gaming platform (e.g., Android, iOS).

The data were organized into a table to facilitate comparisons across the studies based on the usability methods and their effectiveness in identifying usability issues in mobile game UI.

E. Synthesis of Findings

Once the data extraction was complete, the findings from the included studies were analyzed and synthesized to identify common themes, trends, and key findings. This synthesis process involved grouping studies according to the usability evaluation methods for employed. Each method was evaluated for its strengths, weaknesses, and applicability to mobile game UI design. The effectiveness of each method in identifying common usability issues (such as navigation problems, control layout, or feedback mechanisms) was also assessed [41].

F. Evaluation of Methodological Quality

The quality of the selected studies was assessed based on established criteria for systematic reviews, including:

- Study design: The robustness of the study design, including the sample size and the clarity of the evaluation process.
- Methodological rigor: The thoroughness and consistency in applying usability evaluation methods.
- Relevance: The applicability of the study findings to current mobile game UI design practices.
- Transparency: The clarity with which the study reports its findings, including the methodology, results, and conclusions.

- Studies that lacked methodological rigor or transparency were given less weight in the synthesis process.

Studies that lacked methodological rigor or transparency were given less weight in the synthesis process. To ensure methodological rigor, each study was assessed using a modified PRISMA framework adapted for HCI reviews. Potential biases during study selection were mitigated through cross-validation between two independent coders [42].

G. Limitations of the Methodology

While this systematic review aims to provide comprehensive insights into the state of usability evaluation in mobile game UI, there are limitations to the methodology. The review is limited to studies published in English and may exclude valuable insights from non-English research. Additionally, the study only considers studies published in peer-reviewed journals and conferences, potentially overlooking relevant industry reports or grey literature. Finally, the methodology only examines usability evaluation methods from an HCI perspective and does not include other evaluation techniques, such as player-based evaluations or A/B testing [43].

IV. RESULT

A. Growth and Importance of Usability Evaluation in Mobile Games

The mobile gaming industry has grown exponentially, becoming a major player in interactive systems and emphasizing the need for enhanced user experience (UX), particularly through well-designed User Interface (UI) that boost engagement, satisfaction, and retention. Human-Computer Interaction (HCI) plays a crucial role in optimizing mobile game UI by offering usability evaluation methods such as heuristic evaluations, cognitive walkthroughs, and usability testing, which help developers understand and improve how users interact with game interfaces. Despite this, many developers struggle to implement these methods effectively, often resulting in complex games with poor usability and high abandonment rates. While previous research has addressed individual methods, a comprehensive synthesis focusing on their application in mobile game UI design is lacking. This study addresses that gap by systematically reviewing existing usability evaluation techniques, analyzing their strengths, limitations, and real-world applicability to support better UI design in mobile game development.

B. HCI Methods Applied to Mobile Game UI

In the mobile gaming industry, commonly used usability evaluation methods for assessing game UI include heuristic evaluations, cognitive walkthroughs, and usability testing, each offering distinct advantages. Heuristic evaluations rely on established usability principles to identify common issues like poor navigation and ambiguous menu structures, while cognitive walkthroughs simulate player thought processes to uncover obstacles in task completion. Usability testing, the most comprehensive approach, involves observing real users to gather data on performance and satisfaction. When applied

across different game genres, usability testing proves especially effective for action-oriented games due to their real-time interaction demands, whereas cognitive walkthroughs are better suited for narrative-driven adventure games where task progression and problem solving are key. Table I compares these methods across genres, emphasizing their respective strengths in addressing usability challenges such as navigation clarity, feedback quality, and control mapping.

C. UI Design Challenges in Mobile Games

Designing mobile game UI presents inherent challenges due to limited screen space, touch-based interactions, and increasingly complex user behaviors, making intuitive UI design essential for sustaining user engagement and satisfaction. Poorly designed interfaces often frustrate players and disrupt the gameplay experience. As shown in Figure 2, effective UI design is influenced by communication, psychology, core design principles, and responsiveness to user feedback, all of which contribute to smoother navigation and interaction within the game. Heuristic evaluations play a vital role in identifying common usability problems like unclear navigation and poorly placed menus. By applying established usability principles, developers can detect and resolve interface issues, as demonstrated in Table I, leading to more seamless and user-friendly game experiences.

D. Usability Evaluation in Mobile Games: Case Study of Mobile Legends

The case study of Mobile Legends: Bang Bang (MLBB), one of the most popular mobile multiplayer games, demonstrates the application of usability evaluation methods in real-world scenarios. MLBB has undergone several usability assessments using heuristic evaluation and usability testing. The game UI was evaluated based on criteria such as navigation clarity, player control feedback, and visual representation of game data. As demonstrated in Table I, Heuristic 1 (consistent responses to user actions) and Heuristic 6 (intuitive input mapping) were satisfied, showcasing how well the game integrates user-friendly elements into its design.

Additionally, Figure 2 illustrates how usability testing was employed to identify player satisfaction levels in different game modes, revealing areas where players experienced frustration and areas where the interface excelled in supporting gameplay. The usability evaluation of MLBB confirms the importance of integrating both heuristic evaluations and user testing in game development to improve usability and player experience.

E. Key Findings from the Usability Evaluation

The analysis of 34 peer-reviewed studies on mobile game UI reveals key insights into the effectiveness of usability evaluation methods: 62% utilized heuristic evaluation, 50% applied usability testing, and 32% used cognitive walkthroughs. Notably, 25% of reported issues were related to navigational complexity and 18% to inadequate feedback, underscoring both the popularity of certain methods and the persistence of

specific UI challenges. Heuristic evaluation proved effective for identifying common issues early, especially in navigation and interface consistency; cognitive walkthroughs offered deep insights into task-related difficulties in narrative-driven games; and usability testing, though resource-intensive, provided the most comprehensive data on user satisfaction and performance. With the increasing use of AI and AR in mobile games, usability evaluation faces new challenges, requiring more adaptive and context-aware methods to maintain usability in increasingly dynamic gameplay environments.

1) *Heuristic Evaluation Method:* Heuristic evaluation, the most frequently used method in the reviewed studies, proved effective in quickly identifying high-level usability issues, particularly navigation-related problems 25% of which involved unclear menus and confusing user flows in mobile games, consistent with findings. Its strengths lie in efficiency and cost-effectiveness, enabling developers to detect major flaws early, though it may miss context-specific issues in complex interactions, such as those found in adventure games.

Figure 2, which highlights the key aspects for game designers, underscores the importance of understanding the user experience from a holistic perspective. By considering not only usability but also cognitive load and emotional response, designers can create better interfaces that go beyond just usability flaws.

2) *Cognitive Walkthrough Method:* Cognitive walkthroughs are particularly effective for evaluating mobile games with complex task flows, such as adventure games, by simulating user thought processes to uncover issues in decision-making and task sequences for example, successfully applied this method to identify problems like unclear instructions in an adventure game. While valuable for structured, goal-driven gameplay, cognitive walkthroughs are time-consuming and less suitable for fast-paced, action-oriented games. Table I highlights their effectiveness in narrative-rich genres but reduced applicability in games requiring rapid, real-time interactions.

3) *Usability Testing Method:* Usability testing, considered the most comprehensive evaluation method, involves observing real users to gather data on behavior, performance, and satisfaction especially valuable for mobile games with complex controls or multi-touch interactions. Studies showed that poorly placed or unintuitive controls in action games led to lower user satisfaction reinforcing the link between UI complexity and player frustration. Despite its strengths, usability testing is resource-intensive and limited by small sample sizes. As illustrated in Figure 1, top games like PUBG Mobile and Candy Crush benefit from this method to ensure high engagement and identify interface flaws that may impact millions of users.

4) *Comparison of Evaluation Methods:* In Table I, a comparison of the usability evaluation methods used in different game genres is provided. The table clearly shows that heuristic evaluation is the most cost-effective and efficient method, making it suitable for early stages of development. However, it is less effective for identifying more complex usability issues.

Cognitive walkthroughs, while providing detailed insights into task flow and complexity, are more time-consuming and resource-intensive, making them better suited for complex, narrative-driven games. Usability testing, the most comprehensive of all methods, offers the richest data, but it is costly and requires significant resources to conduct.

These findings suggest that a combination of methods is the most effective approach. Heuristic evaluation should be used early in the design process to identify basic usability flaws, while usability testing and cognitive walkthroughs should be used later to refine the interface and ensure that it meets user needs.

V. CONCLUSION

This study systematically examined the use of usability evaluation methods in mobile game UI design, focusing on heuristic evaluation, cognitive walkthroughs, and usability testing. The review emphasized the crucial role of Human Computer Interaction (HCI) in improving user experience in the rapidly growing and increasingly complex mobile gaming industry. While each method has its strengths heuristic evaluation being cost-effective, cognitive walkthroughs offering deep behavioral insights, and usability testing providing rich, real-time data none serve as a universal solution.

Each comes with trade offs in terms of depth, time, and resource requirements. Despite their potential, a significant gap remains in how these methods are integrated into the actual development process. Many developers still struggle to apply user centered design principles effectively, leading to games with complex mechanics but suboptimal usability.

As the industry adopts emerging technologies like AI and AR, traditional evaluation approaches may fall short in addressing the demands of dynamic and adaptive interfaces. This shift underscores the urgent need for evolving usability methods that can keep pace with real-time, technology-driven user experiences.

ACKNOWLEDGMENT

The authors would like to express their sincere gratitude to Universitas Sains dan Teknologi Komputer (STekom) for the institutional support and academic resources provided throughout the research process. Special thanks are also extended to the peer reviewers whose insightful feedback greatly improved the quality and clarity of this paper. This study was made possible through the collaborative effort of the Department of Computer System and the Department of Informatic Engineering, and it reflects our shared commitment to advancing research in Human-Computer Interaction and mobile game development.

REFERENCES

- [1] Y. Gui and F. Zhao, "Personalize mobile game interface design," in *International Conference on Human-Computer Interaction*. Springer, 2024, pp. 22–31.
- [2] H. R. Kadaskar, "Enhancing user experience in mobile application design through gestural interaction: a human-computer interaction perspective," *International Journal of Scientific Research in Modern Science and Technology*, vol. 3, no. 8, pp. 01–06, 2024.

- [3] S. Hong, H. Choi, and H. Kweon, "Medical device based on a virtual reality-based upper limb rehabilitation software: Usability evaluation through cognitive walkthrough," *JMIR Formative Research*, vol. 9, no. 1, p. e68149, 2025.
- [4] A. Alshabib, N. Alakkas, and J. Hassine, "A comparative study of heuristic evaluation and cognitive walkthrough: An e-government usability assessment case study," *Arabian Journal for Science and Engineering*, pp. 1–30, 2025.
- [5] R. Fitria, "Integrating heuristic evaluation and cognitive walkthrough in usability evaluation of mobile application," *Journal of Artificial Intelligence and Engineering Applications (JAIEA)*, vol. 3, no. 3, pp. 667–670, 2024.
- [6] L. Talero-Sarmiento, M. Gonzalez-Capdevila, A. Granollers, H. Lamos-Diaz, and K. Pistili-Rodrigues, "Towards a refined heuristic evaluation: Incorporating hierarchical analysis for weighted usability assessment," *Big Data and Cognitive Computing*, vol. 8, no. 6, p. 69, 2024.
- [7] Q. Aini, D. Manongga, U. Rahardja, I. Sembiring, and Y.-M. Li, "Understanding behavioral intention to use of air quality monitoring solutions with emphasis on technology readiness," *International Journal of Human-Computer Interaction*, pp. 1–21, 2024.
- [8] M. Murod, S. Anhar, D. Andayani, A. Fitriani, and G. Khanna, "Blockchain based intellectual property management enhancing security and transparency in digital entrepreneurship," *Aptisi Transactions on Technopreneurship (ATT)*, vol. 7, no. 1, pp. 240–251, 2025.
- [9] Q. Li, H. Zeng, Z. Peng, and X. Ma, "Understanding players' interaction patterns with mobile game app ui via visualizations," in *Proceedings of the Ninth International Symposium of Chinese CHI*, 2021, pp. 9–21.
- [10] P. A. Suraya, T. Ramadhan, N. Lutfiani, A. Khoirunisa, and U. Rahardja, "Blockchain, information and speculation calculations in indonesia: Recent work," in *2022 10th International Conference on Cyber and IT Service Management (CITSM)*. IEEE, 2022, pp. 1–8.
- [11] T. Schumacher, M. Mortaga, and A. C. Valdez, "A template course for teaching the development of interactive systems to students of human-computer interaction," in *International Conference on Human-Computer Interaction*. Springer, 2024, pp. 211–229.
- [12] F. A. Yaseen, M. S. Ahmed, and M. E. Khudhur, "Enhancing human computer interaction (hci) through machine learning application," in *2024 8th International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT)*. IEEE, 2024, pp. 1–4.
- [13] J. Brewer and M. Romine, "Breaking barriers in mobile game development," *Proceedings of the ACM on Human-Computer Interaction*, vol. 8, no. CHI PLAY, pp. 1–20, 2024.
- [14] H. Ahvenniemi, "Developing an easy-to-use usability-playtesting guide," 2024.
- [15] F. E. Putra, M. Khasanah, and M. R. Anwar, "Optimizing stock accuracy with ai and blockchain for better inventory management," *ADI Journal on Recent Innovation*, vol. 6, no. 2, pp. 190–200, 2025.
- [16] W. A. Social. (2023) The changing world of digital in 2023. We Are Social. Accessed: 2025-07-04. [Online]. Available: <https://wearesocial.com/id/blog/2023/01/the-changing-world-of-digital-in-2023-2/>
- [17] O. Bianchi and H. P. Putro, "Artificial intelligence in environmental monitoring: Predicting and managing climate change impacts," *International Transactions on Artificial Intelligence*, vol. 3, no. 1, pp. 85–96, 2024.
- [18] P. Weichbroth, "Usability testing of mobile applications: A methodological framework," *Applied Sciences*, vol. 14, no. 5, p. 1792, 2024.
- [19] O. Alhadreti, "Assessing the usability of m-health applications: A comparison of usability testing, heuristics evaluation and cognitive walkthrough methods," *International Journal of Advanced Computer Science & Applications*, vol. 15, no. 11, 2024.
- [20] A. Z. Yamani, H. A. Al-Shammare, and M. Baslyman, "Establishing heuristics for improving the usability of gui machine learning tools for novice users," in *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, 2024, pp. 1–19.
- [21] Z. Liu, "Evaluating digitalized visualization interfaces: Integrating visual design elements and analytic hierarchy process," *International Journal of Human-Computer Interaction*, vol. 41, no. 9, pp. 5731–5760, 2025.
- [22] S. J. Saluga, H. Peacock, D. D. Seith, C. C. Boone, Y. Fazeli, R. M. Huynh, J. Luo, Z. Naghi, and R. D. Link, "Inter-twine-d: Combining organic chemistry laboratory and choose-your-own-adventure games," *Journal of Chemical Education*, vol. 99, no. 12, pp. 3964–3974, 2022.
- [23] M. Łączyński, T. Gackowski, and M. Gołębiowski, "Variety or similarity: The typology and frequency of occurrence of certain kinds of puzzles present in puzzle-based gamebooks," *Games and Culture*, p. 15554120251315518, 2025.
- [24] H. Setiyowati, M. A. Harriz, E. Junaedi, N. V. Akbariani, and S. Widodo, "Digitalizing pindang industry with business model canvas for sustainable blue economy," *Aptisi Transactions on Technopreneurship (ATT)*, vol. 7, no. 2, pp. 360–370, 2025.
- [25] Y. Wu, S. Y. Min, S. Prabhumoye, Y. Bisk, R. R. Salakhutdinov, A. Azaria, T. M. Mitchell, and Y. Li, "Spring: Studying papers and reasoning to play games," *Advances in Neural Information Processing Systems*, vol. 36, pp. 22 383–22 687, 2023.
- [26] U. Rahardja, I. D. Hapsari, P. H. Putra, and A. N. Hidayanto, "Technological readiness and its impact on mobile payment usage: A case study of go-pay," *Cogent Engineering*, vol. 10, no. 1, p. 2171566, 2023.
- [27] D. L. Arenas, A. Viduani, and R. B. Araujo, "Therapeutic use of role-playing game (rpg) in mental health: A scoping review," *Simulation & Gaming*, vol. 53, no. 3, pp. 285–311, 2022.
- [28] L. L. Ng, R. S. Azizie, and S. Y. Chew, "Factors influencing esl players' use of vocabulary learning strategies in massively multiplayer online role-playing games (mmorpg)," *The Asia-Pacific Education Researcher*, vol. 31, no. 4, pp. 369–381, 2022.
- [29] G. Chan, Y. Huo, S. Kelly, J. Leung, C. Tisdale, and M. Gullo, "The impact of esports and online video gaming on lifestyle behaviours in youth: A systematic review," *Computers in Human Behavior*, vol. 126, p. 106974, 2022.
- [30] F. Syafariani, M. S. Lola, S. S. S. Abd Mutalib, W. N. F. W. Nasir, A. A. K. A. Hamid, and N. H. Zainuddin, "Leveraging a hybrid machine learning model for enhanced cyberbullying detection," *Aptisi Transactions on Technopreneurship (ATT)*, vol. 7, no. 2, pp. 371–386, 2025.
- [31] J. Döpke, T. Köhler, and L. Tegtmeier, "Are they worth it?—an evaluation of predictions for nba 'fantasy sports,'" *Journal of Economics and Finance*, vol. 48, no. 1, pp. 142–165, 2024.
- [32] A. J. Toth, E. Conroy, and M. J. Campbell, "Beyond action video games: Differences in gameplay and ability preferences among gaming genres," *Entertainment Computing*, vol. 38, p. 100408, 2021.
- [33] L. Rampazzo, "Virtual exchange genres: what can genre theories inform us about the first synchronous session?" *Folios*, no. 60, pp. 35–51, 2024.
- [34] F. Arena, M. Collotta, G. Pau, and F. Termine, "An overview of augmented reality," *Computers*, vol. 11, no. 2, p. 28, 2022.
- [35] S. Nilsson, "Adaptive hint system enhancing user experience in puzzle platformers," <https://urn.kb.se/resolve?urn=urn:nbn:se:umu:diva-226829>, 2024, accessed: 2025-05-04.
- [36] H. Chaalal, M. A. Chemrak, and F. Khatemi, "Ai-driven context classification in mobile computing: methodologies and technologies for enhanced user experience," *Studies in Engineering and Exact Sciences*, vol. 5, no. 2, pp. e9280–e9280, 2024.
- [37] A. Namoun, A. Alrehaili, Z. U. Nisa, H. Almoamari, and A. Tufail, "Predicting the usability of mobile applications using ai tools: the rise of large user interface models, opportunities, and challenges," *Procedia Computer Science*, vol. 238, pp. 671–682, 2024.
- [38] S. Ntoa, "Usability and user experience evaluation in intelligent environments: a review and reappraisal," *International Journal of Human-Computer Interaction*, vol. 41, no. 5, pp. 2829–2858, 2025.
- [39] J. Pi, "Game theory and game mechanics design," in *SHS Web of Conferences*, vol. 188. EDP Sciences, 2024, p. 03020.
- [40] F. A. Irawan, A. S. Setiawati, D. F. W. Permana, L. Aditya, N. Rahesti, and D. S. Ghassani, "Slingshot accuracy in traditional games: What is the ideal grip in shooting?" *Retos: nuevas tendencias en educación física, deporte y recreación*, no. 54, pp. 554–560, 2024.
- [41] J. Siswanto, V. A. Goeltom, I. N. Hikam, E. A. Lisangan, and A. Fitriani, "Market trend analysis and data-based decision making in increasing business competitiveness," *Sundara Advanced Research on Artificial Intelligence*, vol. 1, no. 1, pp. 1–8, 2025.
- [42] L. H. Sheng, L. T. Yong, and L. Zhiqiang, "Interaction design for mobile gaming development: A systematic literature review," in *2024 IEEE International Conference on Computing (ICOCO)*. IEEE, 2024, pp. 171–176.
- [43] A. Wali, S. Mahamad, and S. Sulaiman, "Task automation intelligent agents: A review," *Future Internet*, vol. 15, no. 6, p. 196, 2023.